Tasks 1A–4 and Extensions: Exploratory Data Analysis and Modelling of Wine Quality

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Overview

This document presents the full exploratory data analysis (EDA) and modelling tasks conducted as part of the wine quality prediction project. Both red and white wine datasets are analysed. Relationships between alcohol, residual sugar, and quality are explored, followed by feature engineering and classification/regression modelling. Extensions investigate interactions between alcohol and sweetness.

Task 1A: Initial Data Analysis

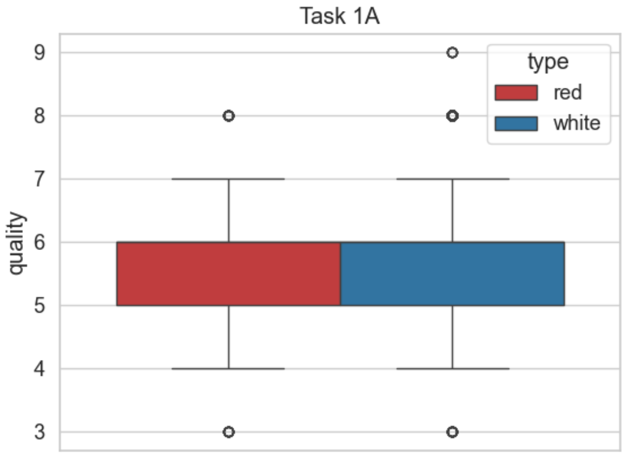

Figure 1: Boxplot of Quality Scores in Red and White Wine Datasets

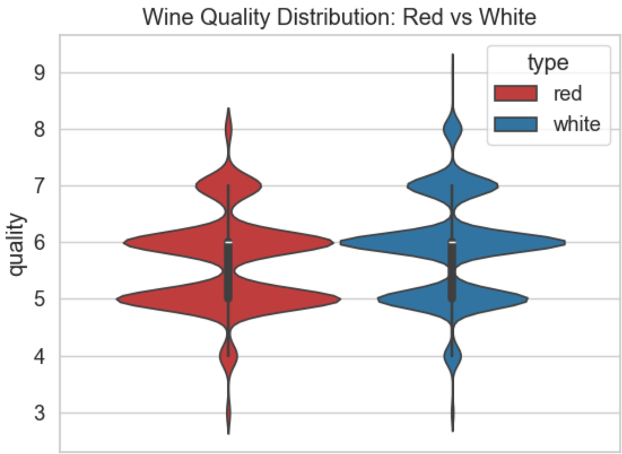

Figure 2: Violin Plot: Red vs White Wine Quality Distribution

Interpretation: Most red and white wines are rated between 5 and 6, but white wines have a wider spread. This may reflect differences in production or grape characteristics.

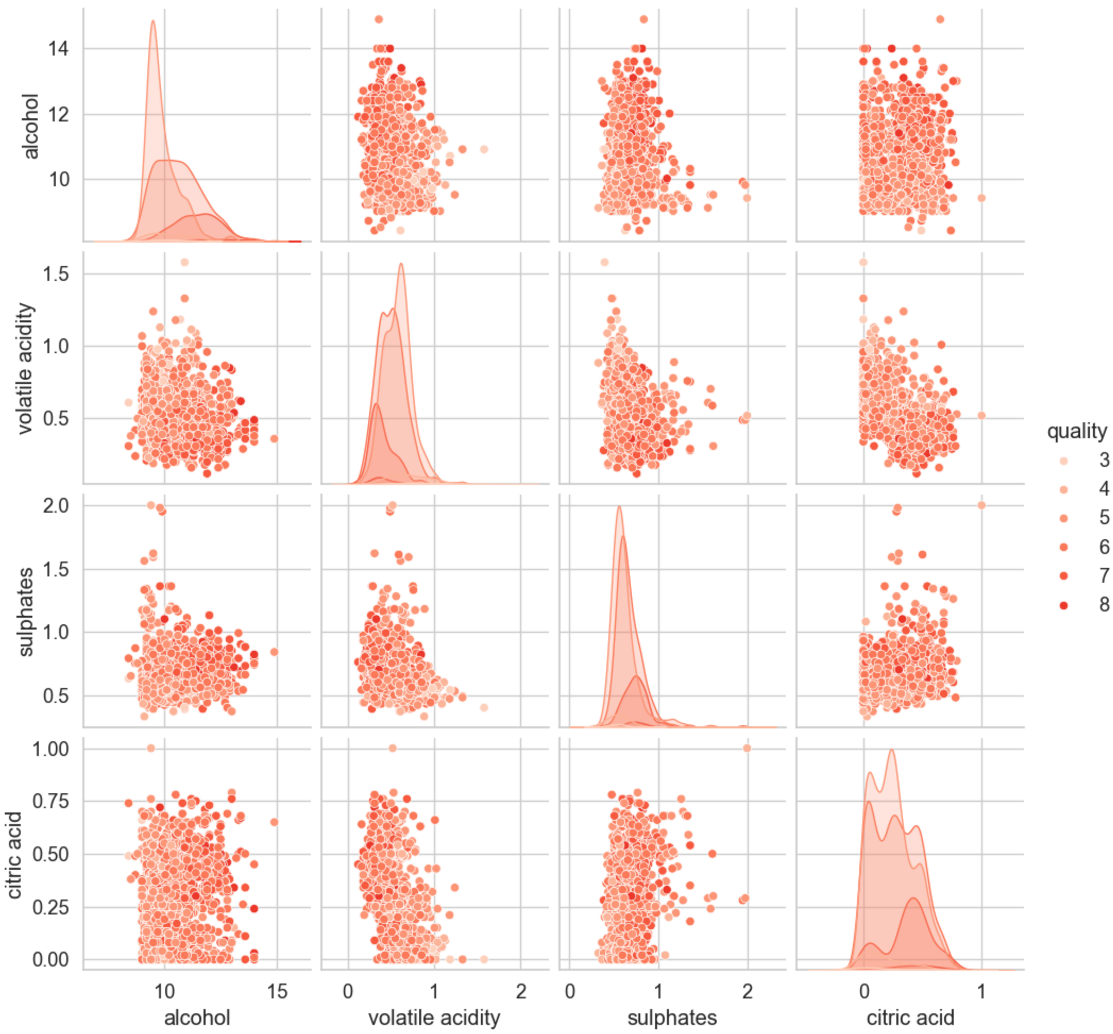

Figure 3: Red Wine: Scatterplot Matrix of Selected Features by Quality

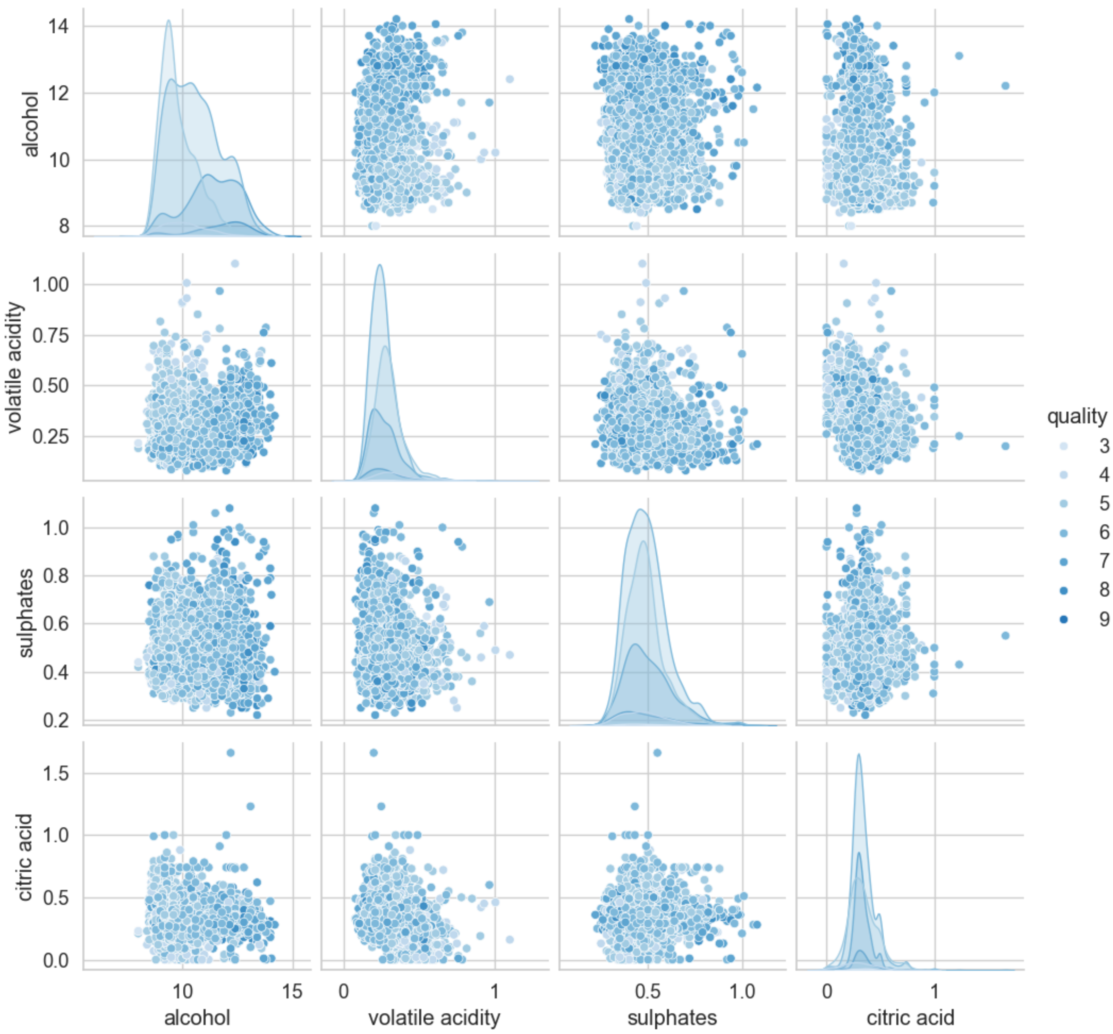

Figure 4: White Wine: Scatterplot Matrix of Selected Features by Quality

Task 1B: Alcohol Content and Quality

Alcohol content was categorised as `low`, `mid`, or `high` based on the mean and standard deviation.

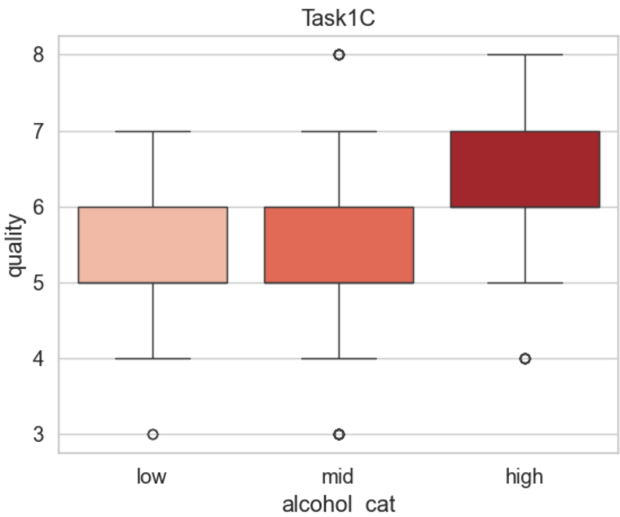

Figure 5: Red Wine: Quality by Alcohol Category

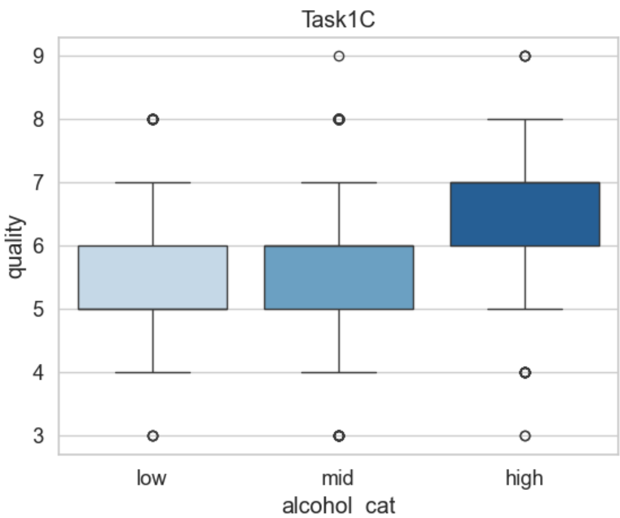

Figure 6: White Wine: Quality by Alcohol Category

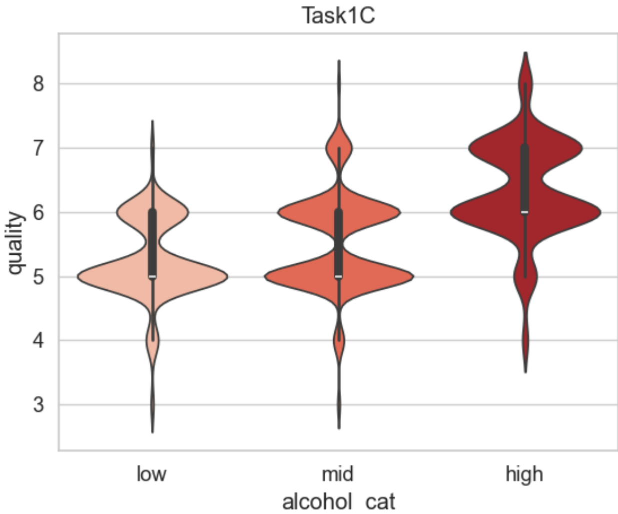

Figure 7: Red Wine: Violin Plot of Quality by Alcohol Category

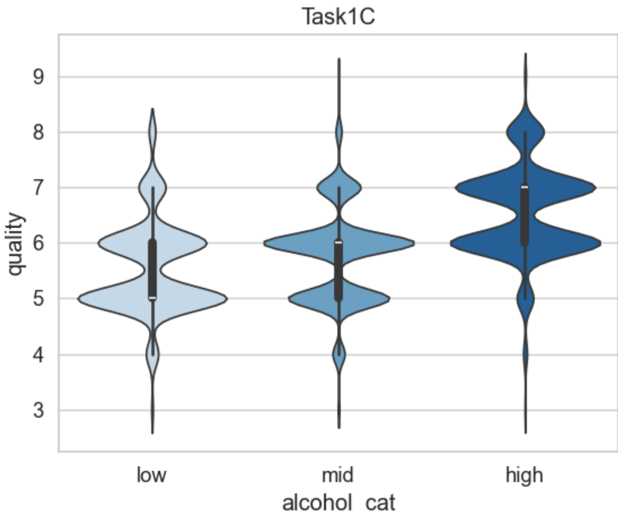

Figure 8: White Wine: Violin Plot of Quality by Alcohol Category

Conclusion: Alcohol is positively associated with quality in both datasets, with red wine showing a clearer pattern.

Task 1C: Residual Sugar and Sweetness

The feature `isSweet` was created by splitting at the median of residual sugar.

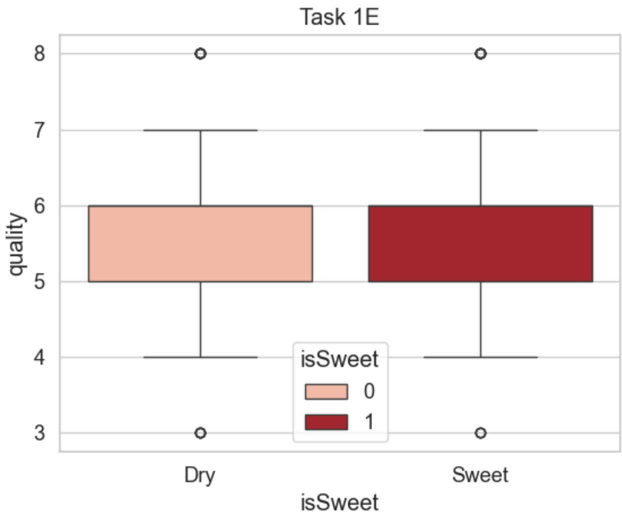

Figure 9: Red Wine: Quality by Sweetness (`isSweet`)

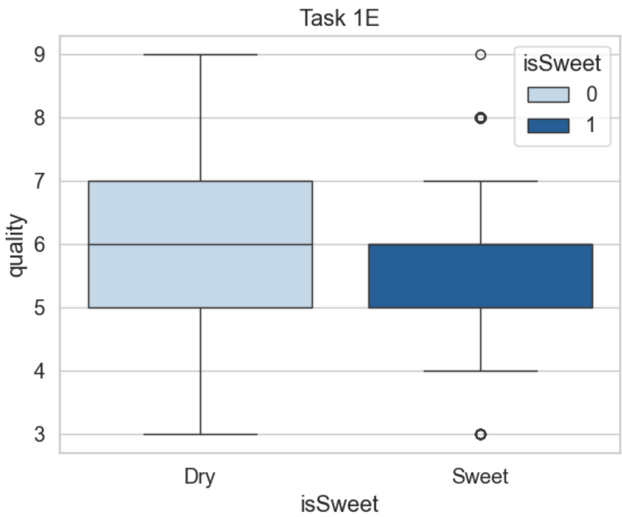

Figure 10: White Wine: Quality by Sweetness (`isSweet`)

Figure 11: Red Wine: Violin Plot of Quality by Sweetness

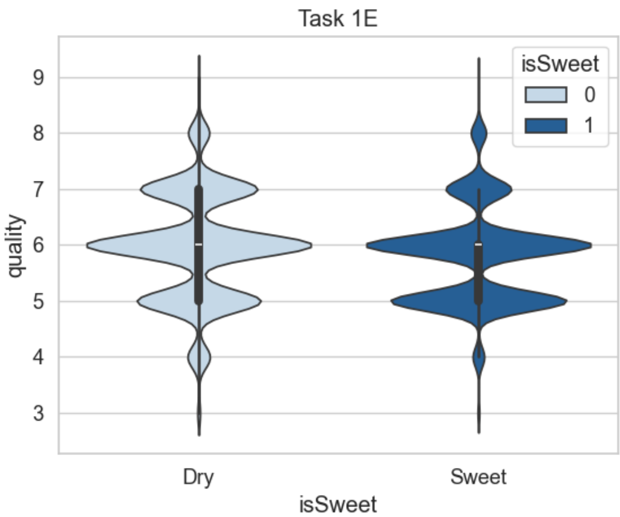

Figure 12: White Wine: Violin Plot of Quality by Sweetness

Conclusion: Sweetness has a weak negative effect on quality, especially in white wine.

Extension 1: Alcohol & Sweetness Interaction

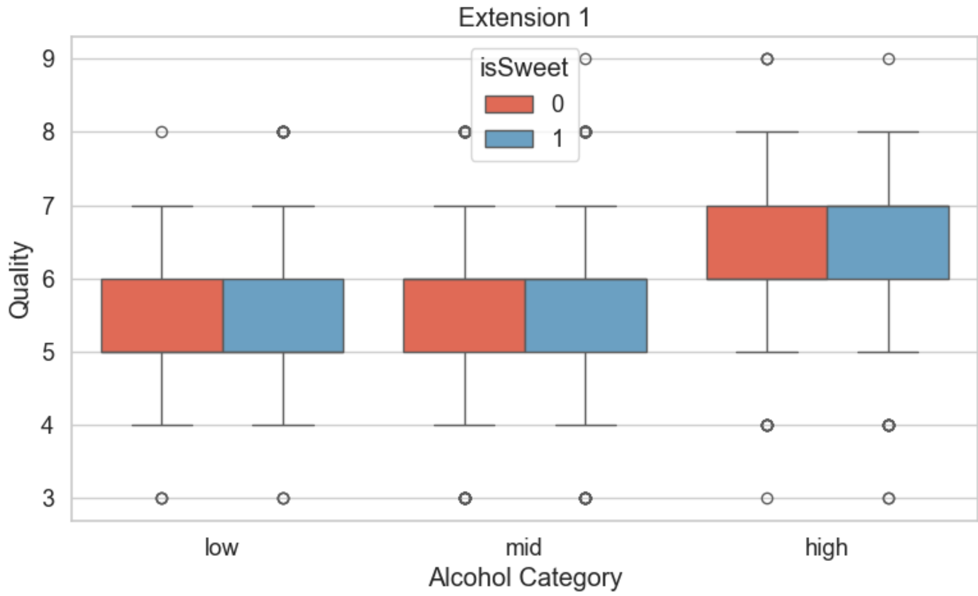

Figure 13: All Wines: Quality by Alcohol and Sweetness

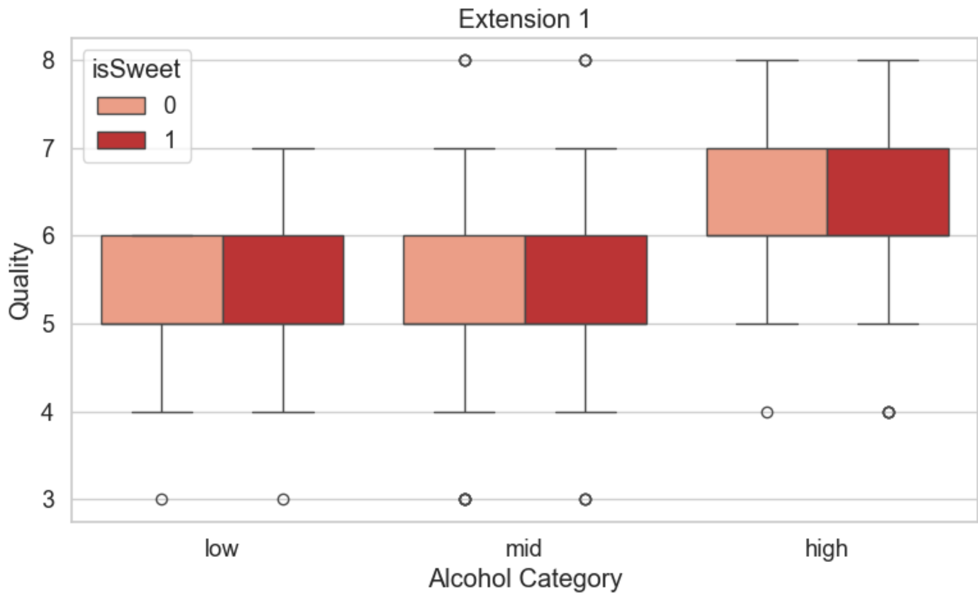

Figure 14: Red Wine: Quality by Alcohol and Sweetness

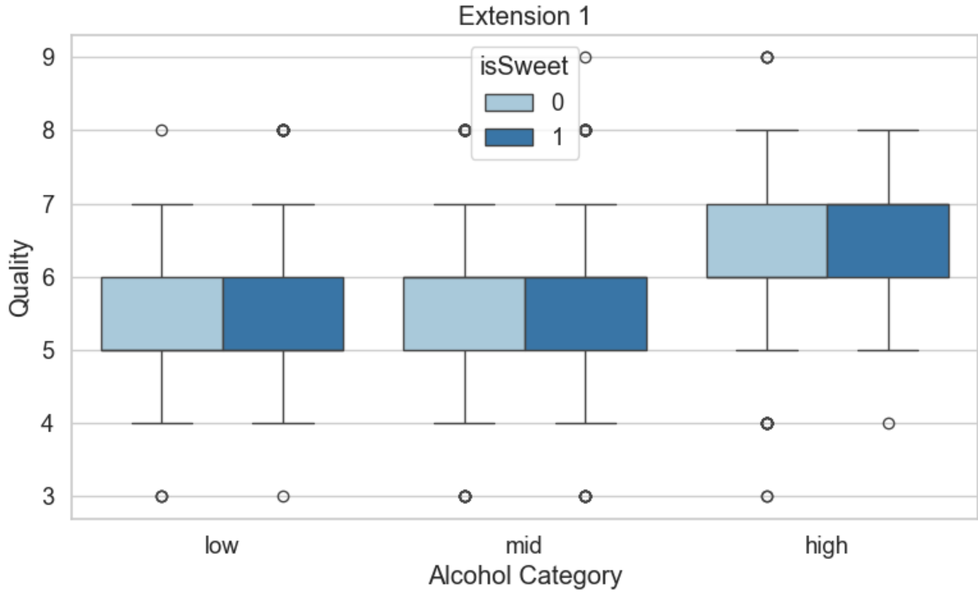

Figure 15: White Wine: Quality by Alcohol and Sweetness

Interpretation: Quality increases with alcohol regardless of sweetness, but dry wines tend to benefit more. Sweetness reduces average quality in high-alcohol white wines.

Task 2: Correlation Analysis

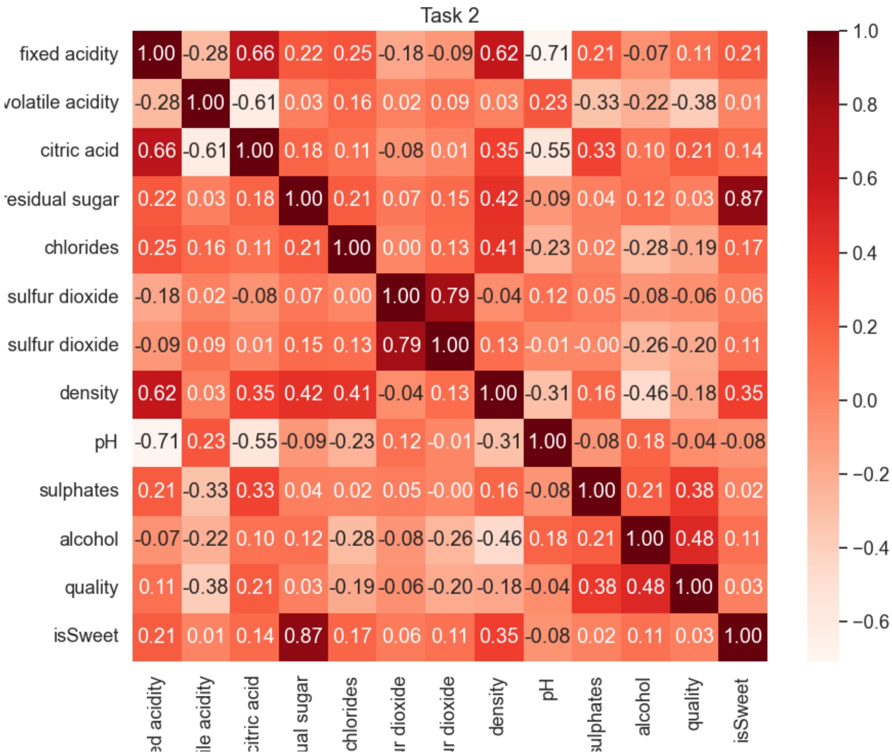

Figure 16: Red Wine: Pearson Correlation Matrix

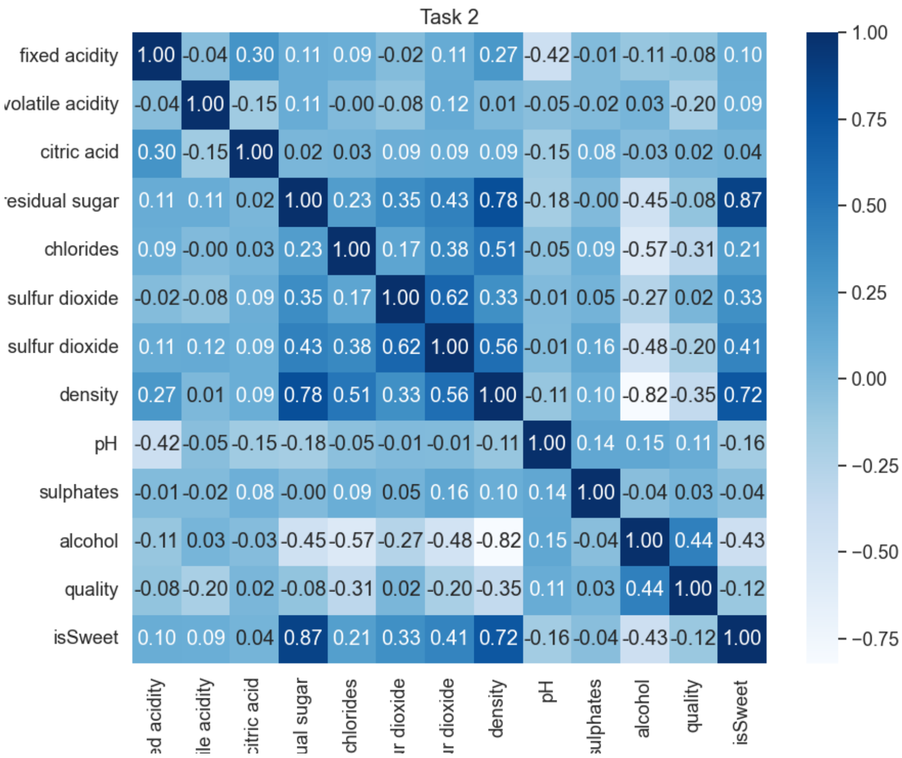

Figure 17: White Wine: Pearson Correlation Matrix

Task 3: Binary Classification

Figure 18: Red Wine: Cross-Validation Accuracy Across Thresholds

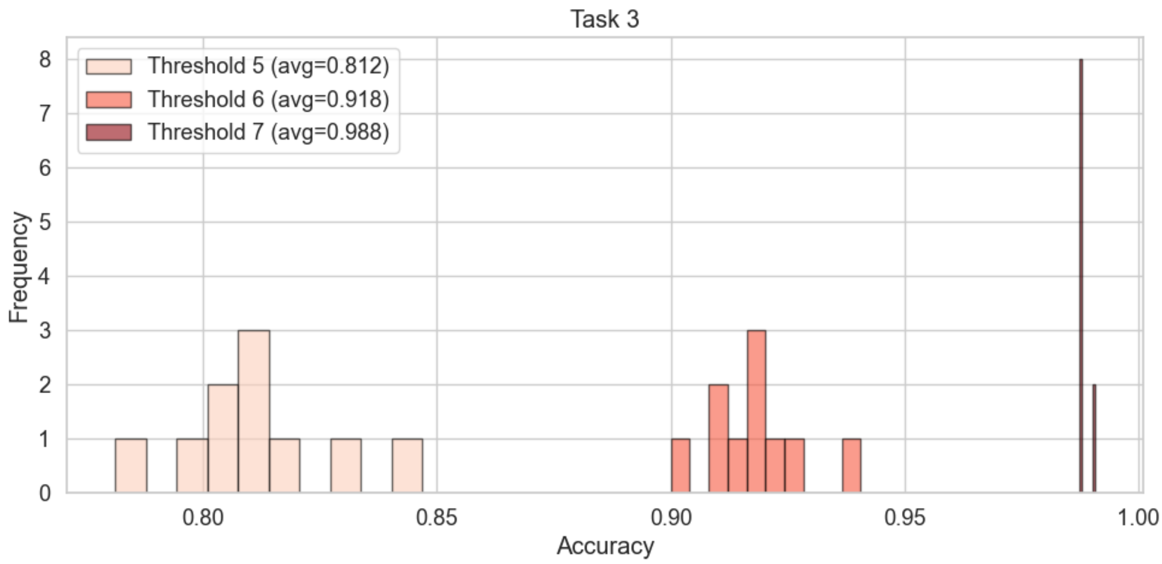

Figure 19: Red Wine: Test Set Accuracy Distribution

Conclusion: Threshold 6 yielded the best balance of performance and class distribution.

Task 4: Regression and ROC Evaluation

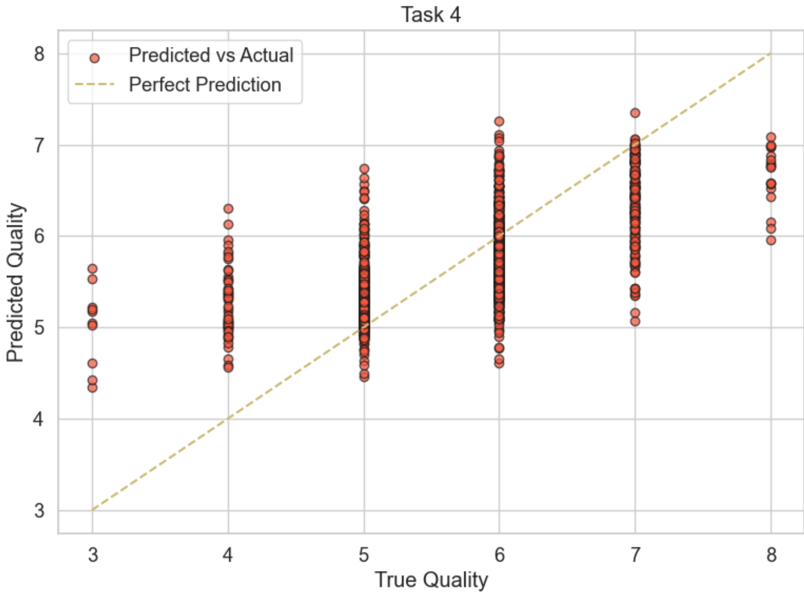

Figure 20: Red Wine: Predicted vs Actual Quality (Regression)

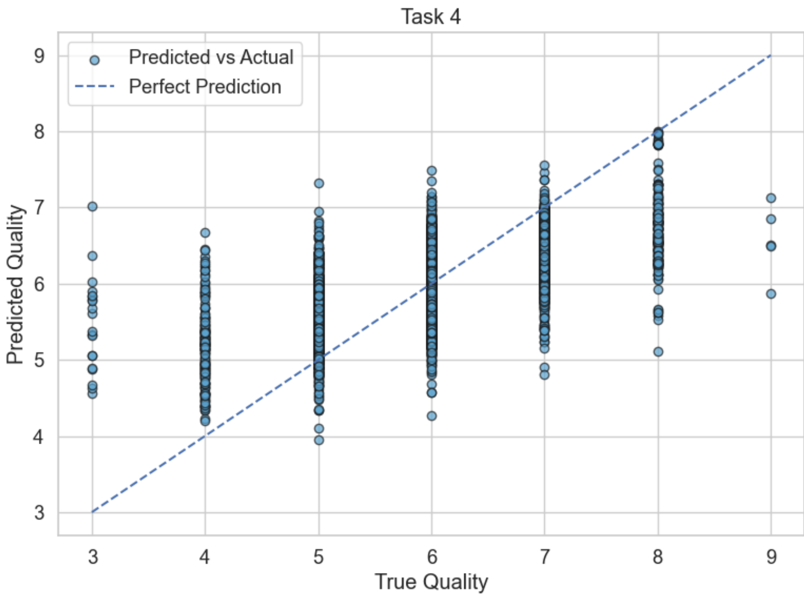

Figure 21: White Wine: Predicted vs Actual Quality (Regression)

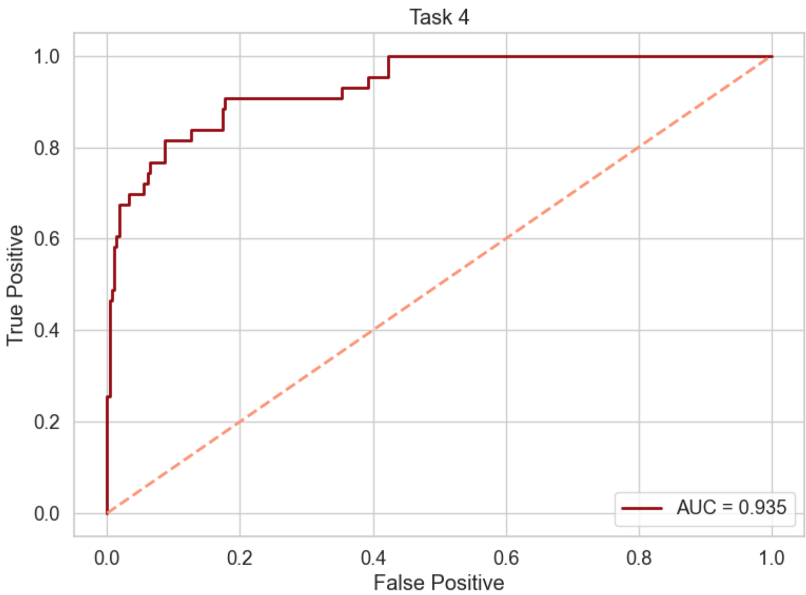

Figure 22: Red Wine: ROC Curve for Threshold 6

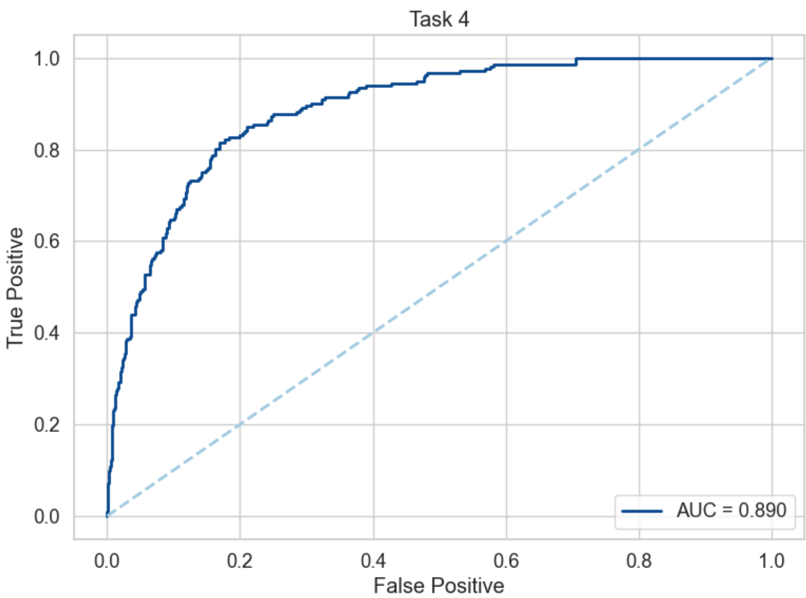

Figure 23: White Wine: ROC Curve for Threshold 6